Substitution Method Analysis

CSA0606-Design and Analysis of Algorithms

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Introduction

What is the Substitution Method?

- Mathematical technique for solving recurrence relations
- Also known as the 'guess and verify' method
- Fundamental for analyzing algorithm complexity
- Works by substituting the recurrence into itself repeatedly
- Helps identify patterns and derive closed-form solutions

Problem Statement

Recurrence Relation

$$T(n) = T(n-1) + n$$

Base Case

- $T(1) = 1$

Objective

- Derive the closed-form solution and determine the time complexity

- **The given recurrence relation is $T(n) = T(n - 1) + n$**, with the base case **$T(1) = 1$**.
- **The objective is to solve the recurrence** using the **Substitution Method**.
- **The recurrence is expanded step by step** until the base case is reached.
- **A closed-form solution is derived** to represent the recurrence without recursion.
- **The final goal is to determine the time complexity** of the recurrence relation after obtaining the closed-form solution.

Algorithm / Pseudocode

Substitution Method Approach

```
Algorithm: Substitution Method
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Step 1: Write the recurrence relation
        T(n) = T(n-1) + n

Step 2: Substitute T(n-1) recursively
        T(n) = T(n-2) + (n-1) + n

Step 3: Continue substitution k times
        T(n) = T(n-k) + (n-k+1) + ... + (n-1) + n

Step 4: Expand until base case is reached
        When n-k = 1, we have performed (n-1) substitutions

Step 5: Use summation formula
        Sum of i from 1 to n = n(n+1)/2

Step 6: Derive closed form and identify complexity
```

Step-by-Step Expansion

Substitution Method - Detailed Analysis

Step 1 $T(n) = T(n-1) + n$

Step 2 $= [T(n-2) + (n-1)] + n = T(n-2) + (n-1) + n$

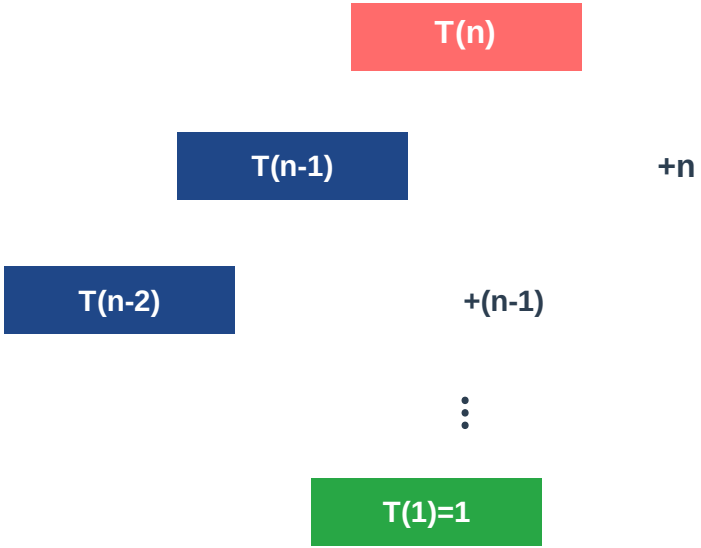
Step 3 $= T(n-3) + (n-2) + (n-1) + n$

Step 4 $= T(n-k) + (n-k+1) + \ldots + (n-1) + n$

Base Case: When $n-k = 1 \Longrightarrow k = n-1$
$T(1) = 1$, so after $(n-1)$ substitutions, we reach the base case

Architecture Diagram

Recursive Decomposition Tree



Depth = n-1 levels | Cost at each level increases

- **The diagram starts with T(n)**, which represents the original problem.
- **Each step calls T(n-1)**, reducing the problem size by 1.
- **At every level, the current value (+n, +(n-1), etc.) is added** to the result.
- **The recursion stops at T(1) = 1**, which is the base case.
- **The diagram shows how all the values are added together** to get the final answer **T(n) = n(n+1)/2**.

Implementation (Code)

Python Implementation

```
# Recursive Implementation
def T_recursive(n):
    if n == 1:
        return 1
    return T_recursive(n-1) + n

# Closed-form Solution
def T_closed_form(n):
    return n * (n + 1) // 2

# Example Usage
for n in range(1, 8):
    recursive = T_recursive(n)
    closed = T_closed_form(n)
    print(f"T({n}) = {recursive}")
```

- **Recursive Function** T_recursive(n) solves the recurrence relation **T(n) = T(n-1) + n** by calling itself until the base case is reached.
- **Base Case** if n == 1: return 1
- Stops the recursion and provides the initial value **T(1) = 1**.
- **Recursive Step** return T_recursive(n-1) + n
- Computes the previous result and adds the current value of **n**, building the sum step by step.
- **Closed-Form Solution** T_closed_form(n) uses the formula **n(n+1)/2** to calculate the result directly without recursion, making it more efficient.
- **Output Verification** The for loop runs from **n = 1 to 7**, computes the result using both recursive and closed-form methods, and verifies that both produce the same values.

Mathematical Derivation

From Substitution to Closed Form

After (n-1) substitutions with base case T(1) = 1:

$$T(n) = T(1) + 2 + 3 + \ldots + (n-1) + n$$

$$T(n) = 1 + \sum_{i=2}^{n} i$$

Using the formula: $\sum_{i=1}^{n} i = n(n+1)/2$

$$T(n) = \sum_{i=1}^{n} i = n(n+1)/2$$

Closed Form: $T(n) = n(n+1)/2 = (n^2 + n)/2$

Output Analysis

Computed Values for Various n

n	T(n) = n(n+1)/2	Verification
1	1	1×2/2 = 1
2	3	2×3/2 = 3
3	6	3×4/2 = 6
5	15	5×6/2 = 15
10	55	10×11/2 = 55
100	5050	100×101/2 = 5050

Complexity Analysis

Time & Space Complexity

Time Complexity

$O(n^2)$

Since $T(n) = n(n+1)/2 = (n^2 + n)/2$, the leading term is n^2, making this a quadratic complexity.

Space Complexity

$O(n)$

Recursive call stack depth is n due to (n-1) recursive calls before reaching the base case.

Efficiency: Linear time algorithms are preferred over quadratic ones. For large n, $O(n^2)$ becomes computationally expensive.

Conclusion

- The Substitution Method is a powerful technique for solving recurrence relations
- Through step-by-step expansion, we derived the closed form: $T(n) = n(n+1)/2$
- The algorithm exhibits quadratic time complexity $O(n^2)$
- Space complexity of $O(n)$ is due to recursive call stack
- This analysis demonstrates why iterative solutions are preferred in practice
- Understanding complexity helps in algorithm selection and optimization

References

Key Resources

- [1] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to Algorithms (3rd ed.). MIT Press.
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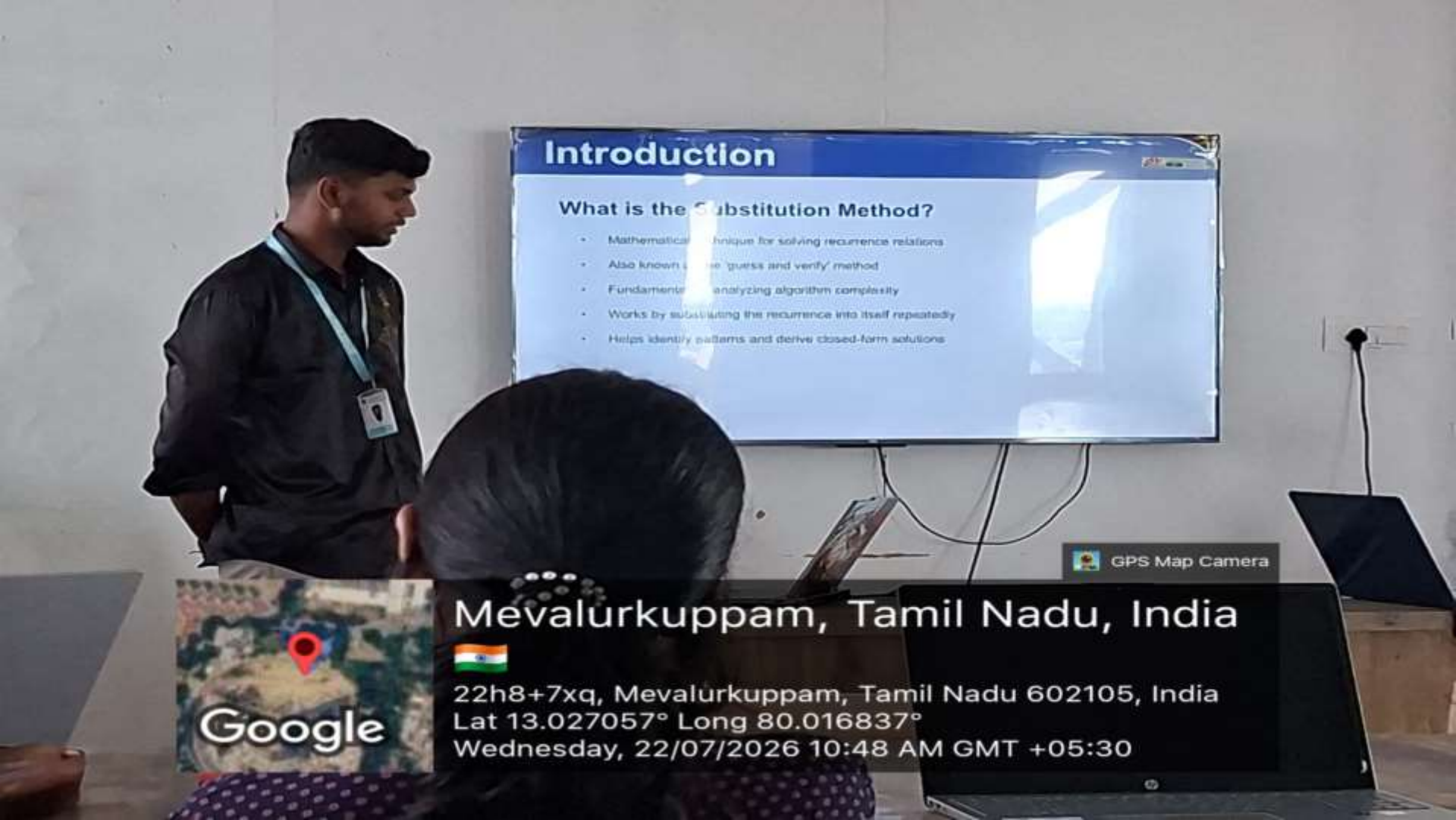
- [3] Sedgewick, R., & Wayne, K. (2011). Algorithms (4th ed.). Addison-Wesley.
- [4] Rosen, K. H. (2011). Discrete Mathematics and Its Applications (7th ed.). McGraw-Hill.

Thank You!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